Prediction And Falsification Protocols From Finite-Capacity Causal Networks

Abraham Greenman

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Abstract

This paper closes the Paper 15 prediction-and-falsification protocol theorem as a conditional, finite, network-native interface theorem downstream of the closed Paper 14 discriminating benchmarks certificate. The result constructs finite protocol records, target and regime descriptors, threshold and rejection-condition descriptors, Paper 14 compatibility rows, stability and reproducibility descriptors, and a fail-closed no-hidden-promotion audit. The final certificate closes only this finite protocol-structure theorem. It does not assert protocol recovery, benchmark success, prediction success, falsification success, physical validation, empirical adequacy, physical promotion, physical nature realization, or a unified field theory.

1 Scope

Paper 15 starts from the frozen Paper 14 conditional discriminating benchmarks certificate:

`paper14_dbm008_final_conditional_certificate_closes_discriminating_benchmarks_theorem`.

The Paper 14 frozen commit is:

`ad4f876a1699874cd6efd7fe73d318e64f5bbe19`.

The theorem closed here is intentionally local. It says that prediction and falsification protocols can be represented as finite, auditable records in the finite-capacity causal-network theorem ladder. It does not say that any prediction has succeeded, any alternative has been falsified, any benchmark has succeeded, or any physical theory has been validated.

2 Claim Boundary

The following claims remain false in this paper:

- protocol recovery;
- benchmark success;
- prediction success;
- falsification success;
- physical promotion;
- physical validation;
- empirical adequacy;
- observed-catalog recovery;

- simulation-only promotion;
- fit-only calibration;
- physical nature realization;
- unified field theory.

All constructions below are finite protocol-interface constructions only.

3 Protocol Records

3.1 Finite protocol row

A Paper 15 protocol row is a finite tuple

$$P = (i, t, o, r, \theta, \rho, a, b, c),$$

where:

- i is a bounded protocol identifier;
- t is a bounded prediction-target label;
- o is a bounded observable label;
- r is a bounded regime label;
- θ is a bounded threshold label;
- ρ is a bounded rejection-condition label;
- a is a bounded audit-status descriptor;
- b is a Paper 14 compatibility reference;
- c is the non-promotion claim boundary.

The row closes 'PFP-002' exactly when the schema is finite, the labels are bounded, the row is auditable, Paper 14 compatibility is referenced only as schema alignment, and the claim boundary contains no success, recovery, validation, promotion, physical-nature, or unified-field claim.

3.2 Target, observable, and regime descriptors

'PFP-003' refines a closed protocol row with finite descriptors for prediction target, observable, and regime. These descriptors are selection criteria only. They do not recover an observed catalog and do not import benchmark success, prediction success, falsification success, validation, empirical adequacy, physical promotion, or unified-field promotion.

3.3 Threshold and rejection-condition descriptors

'PFP-004' adds finite threshold and rejection-condition descriptors. The threshold is a finite criterion label. The rejection condition is a predeclared protocol rule. Closing this rung does not mean that any rejection condition has fired or that any alternative has been falsified.

3.4 Paper 14 compatibility

'PFP-005' binds the protocol stack to the frozen Paper 14 commit, formal endpoint, and final certificate as finite schema alignment:

`Paper14DiscriminatingBenchmarksTheoremContract.closed`.

This compatibility is not benchmark success and is not prediction or falsification success.

3.5 Stability and reproducibility

'PFP-006' defines finite descriptors for protocol stability, coarse-graining, and reproducibility. Reproducibility means compatibility of finite records under preserved labels, audit rows, and claim boundaries. It is not reproduced empirical success, physical validation, benchmark success, prediction success, or falsification success.

3.6 No-hidden-import audit

'PFP-007' is a fail-closed audit. It closes only when every route that could silently import recovery, benchmark success, prediction success, falsification success, validation, empirical adequacy, observed-catalog recovery, simulation-only promotion, fit-only calibration, physical-nature realization, or unified-field promotion remains explicitly blocked.

4 Theorem

Paper 15 Prediction And Falsification Protocols Theorem. Assume the closed upstream Paper 14 conditional discriminating benchmarks certificate and the closed Paper 15 rungs 'PFP-001' through 'PFP-007'. Then there exists a finite, network-native prediction-and-falsification protocol interface whose rows specify bounded protocol identifiers, prediction targets, observables, regimes, thresholds, rejection conditions, Paper 14 compatibility references, stability and reproducibility descriptors, and audit statuses. The interface is closed only as conditional finite protocol structure and preserves the no-recovery, no-success, no-validation, no-promotion, no-physical-nature, and no-unified-field claim boundary.

5 Proof

The Lean formalization defines contracts for each rung:

- 'PFP001UpstreamBindingContract';
- 'PFP002FiniteProtocolRecordContract';
- 'PFP003PredictionTargetRegimeContract';
- 'PFP004FalsificationThresholdContract';
- 'PFP005Paper14BenchmarkCompatibilityContract';
- 'PFP006StabilityReproducibilityContract';
- 'PFP007NoHiddenPromotionValidationSuccessAuditContract';
- 'PFP008FinalConditionalCertificateContract'.

Each contract has a 'closed' predicate that is the conjunction of its finite construction requirements and its claim-boundary guards. For 'PFP-001' through 'PFP-007', canonical contracts close their local rung while the Paper 15 theorem contract remains open because the final certificate field is still false.

The final certificate 'PFP-008' consumes the closed canonical contracts for 'PFP-001' through 'PFP-007', adds the final conditional-certificate fields, and preserves every no-claim guard. The formal endpoint is:

paper15_pfp008_final_conditional_certificate_
closes_prediction_falsification_protocols_theorem

This theorem proves:

`Paper15PredictionFalsificationProtocolsTheoremContract.closed`

for the final canonical Paper 15 theorem contract.

6 Audit Surface

The Rust audit surface mirrors the Lean contracts with finite record types and regression tests. The final Rust certificate closes only when:

- the upstream Paper 1–14 binding closes without physical claims;
- the 'PFP-002' through 'PFP-007' audit records close;
- the final certificate is explicitly conditional;
- all claim boundaries remain false.

The regression suite also checks these failure cases:

- overlong labels, unlinked records, and bad Paper 14 bindings;
- benchmark-success, prediction-success, and falsification-success leakage;
- empirical-validation leakage and unblocked hidden imports;
- nonconditional final certificates and unified-field promotion.

7 Reproducibility

The repository checks are:

```
make test-fast
make lean-build
```

The final state is recorded in 'GPD/state.json' with:

```
"active_obligation": "NONE"
"prediction_and_falsification_protocols_theorem_closed": true
"paper15_canonical_proof":
  "paper15_pfp008_final_conditional_certificate_"
  "closes_prediction_falsification_protocols_theorem"
```

8 Conclusion

Paper 15 closes the finite prediction-and-falsification protocol interface theorem conditionally. The result is a theorem about finite protocol records and their audit boundaries. It is not evidence that predictions succeeded, not evidence that alternatives were falsified, not a physical validation result, and not a unified field theory claim.